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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting makes it harder to search for items
- c. Sorting decreases the efficiency of selection operations
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **3**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Using the divide-and-conquer approach
- c. Sorting data in a non-sequential manner
- d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Bubble Sort
c. Insertion Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is difficult to implement
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Merge sort
c. Heap sort
d. Quick sort

The correct answer is:
Merge sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It divides the list into sublists
- b. It bubbles up the largest element in each pass
- c. It uses the heap data structure
- d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Backtracking
- b. Divide and conquer
- c. Greedy method
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Finding a specific element in a list
- c. Removing duplicates from a list
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It compares adjacent elements and swaps them if necessary
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **11**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It is less efficient than bubble sort
c. It sorts data using a single pass
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It only works with integer arrays
c. It sorts data out of the box efficiently
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Sorting
d. Searching

The correct answer is: Sorting

Question **14**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 80,150,200,470,40,90,300,400,70,120
c. 40,80,90,150,200,300,400,470,70,120
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By using the bubble-up method
c. By sorting data in a single pass
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them